Spatiotemporal Forecasting of Hourly PM2.5 in Delhi-NCR: Does a Wind-Aware Graph Beat a Static Distance Graph?

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Abstract

We forecast hourly PM2.5 at 42 monitoring stations across Delhi-NCR over 2019–2025 and ask whether a wind-aware graph provides predictive gains over a static distance-based graph. On held-out 2024–2025 data the best neural model (per-station STGCN without a graph, predicting changes from the last observation) reaches MAE 32.61 μg/m^3, against 44.03 for ridge regression on the last 24 lags and 46.79 for persistence. The graph comparison is judged with a paired bootstrap over the 105 test weeks rather than a per-seed standard deviation. For Graph WaveNet the wind-aware graph beats the static distance graph (33.76 vs. 34.39 μg/m^3, difference 0.63, 95% CI [0.36, 0.91]); for STGCN the two are indistinguishable (0.12, CI [−0.15, 0.41]). In both architectures, dropping the distance graph is significantly better than keeping it; dropping the wind-aware graph is also better, significantly so for STGCN (no graph wins by 0.45, CI [0.15, 0.79]) and not significantly for Graph WaveNet (0.19, CI [−0.15, 0.55]). The graph learned end-to-end (Graph WaveNet's adaptive adjacency) and the correlation graph are the two worst configurations; the correlation graph reaches 35.03 μg/m^3 (+1.46 vs. no graph, CI [0.70, 2.29]). A network-mean diagnostic shows that handing the model the network-wide average explicitly gains essentially nothing (0.01 μg/m^3), so inter-station mixing is not delivering a useful regional signal. The consistent reading is that the per-station autoregressive signal dominates and inter-station mixing is at best a weak positive and at worst a net cost. Whether the mixing is harmful or just redundant is a hypothesis consistent with the data, not a demonstrated mechanism.

1 Introduction

Air pollution in Delhi-NCR is a severe public health crisis. Accurate hourly PM2.5 forecasts at station level could enable targeted interventions. Recent literature proposes spatiotemporal graph neural networks (STGNNs) for air quality forecasting, claiming that graph structure—whether static (distance, correlation) or wind-aware—captures pollutant transport and improves predictions. This work tests that claim on a frozen, real-world dataset spanning 2019–2025 across Delhi-NCR monitoring stations, with the graph hypothesis isolated behind a controlled ablation and judged with a paired bootstrap rather than a per-seed standard deviation.

2 Related work

Spatiotemporal graph networks for forecasting were popularised by the diffusion convolutional recurrent neural network of Li et al.[3], which placed a recurrent model on a diffusion graph over road sensors and showed that capturing spatial dependence improved over a convolutional baseline. Yu et al.[2] separated spatial and temporal operators into a spectral graph convolution followed by gated temporal convolutions; the STGCN we train here is a GRU-based variant of that architecture. Graph WaveNet [1] introduced dilated causal convolutions with a fixed or

learned adjacency. On the air-quality side, spatiotemporal graph networks have been applied to PM2.5 forecasting, typically with a distance-threshold graph, sometimes with a wind-aware graph, reporting gains over graph-free baselines [4]. Two aspects of that literature motivate this study. First, the reported gains are usually measured against weak baselines, so a graph-free sequence model can look competitive. Second, the comparison is rarely made with enough repetitions to resolve the small differences that graph structure actually produces.

3 Data

3.1 Source and coverage

Hourly PM2.5 comes from the CPCB continuous monitoring network, obtained as 15-minute records for 2009–2025 from the `Vonter/india-cpcb-aqi` release archive. After resampling to hourly and dropping stations below 70% real-observation coverage in any split, the modelling set is 42 stations with approximately 61,345 hourly timestamps over 2019-01-01 to 2025-12-31. Station meteorology (temperature, relative humidity, wind speed and direction, rainfall) is measured at the stations themselves.

3.2 Provenance and the observation mask

Table 1 lists every source. No synthetic data is used. The build fails if any station falls below 70% real observations in any split.

Table 1: Data sources. No synthetic data is used anywhere.

what	source	accessed via	licence
hourly PM2.5 + station meteorology	CPCB continuous monitoring, 15-minute records	https://github.com/Vonter/india-cpcb-aqi (release archive)	ODbL as published by CPCB
ERA5 reanalysis wind (ablation only)	ERA5 via Open-Meteo archive API	https://archive-api.open-meteo.com/v1/archive	Copernicus licence (free, attribution required)
station metadata (lat/lon, city)	CPCB station list + XKDR registry	`data/raw/station_coords.csv`	public metadata

The target is never imputed. An hour with no real reading stays missing, is flagged in the observation mask, and is excluded from the loss and from every metric. Table 2 gives the resulting coverage; 6.5% of the test split is missing, and the metrics reported throughout are computed on the observed remainder.

Table 2: Observation-mask audit. The target is never imputed.

split	hours	observed	missing	mean PM2.5	frac >300
train	35,041	90.80%	9.20%	102.0	4.82%
val	8,760	92.32%	7.68%	98.0	4.23%
test	17,521	93.52%	6.48%	97.5	4.15%

3.3 Splits and leakage control

Splits are chronological and non-overlapping: train 2019–2022 (34,922 windows), validation 2023 (8,641 windows), test 2024–2025 (17,402 windows). The split window counts sum to 60,965; the

23-hour gap that separates the test block from validation accounts for the difference from the 61,345 total hours. Every window's inputs *and* targets lie wholly inside its own split, verified programmatically. The train and validation blocks are contiguous; a 23-hour gap separates the test block from validation. Normalisation ($\log(1+x)$ then z-score) is fitted on the training split only, every graph is constructed from training-split information only, and imputed hours carry a mask honoured in both the loss and the metrics.

4 Methods

4.1 Baselines

Persistence repeats the last observed value. **Seasonal naive** repeats the value observed 24 hours before the target; this is a far stronger baseline than plain persistence because Delhi's PM2.5 has a large diurnal cycle. **Hour-of-week average** takes, for each station and hour of the week, the mean observed PM2.5 over the training split. **Ridge** fits a per-station linear map from the last 24 hourly values to each horizon on the training split only.

4.2 Models

LSTM (univariate, shared weights per station; multivariate, stations as channels), a temporal Transformer, STGCN (spectral graph convolution + GRU), and Graph WaveNet (dilated causal convolutions with fixed or learned adjacency). Fifteen configurations in total.

Two design choices matter more than architecture. First, every network predicts the *change* from the last observed value and adds that value back at the output; the output layer is zero-initialised, so each model *starts* at Persistence. Second, each model carries a linear autoregressive skip from the last 24 lags to the 48 horizons, so the network's contribution is measured relative to a strong linear floor.

Training details. All models are trained with a masked Huber loss on the change from the last observation ($\delta = 1.0$), Adam (initial LR 10^{-3}, cosine decay), batch size 32, early stopping on validation MAE (patience 3). The input window is 72 hours and the output horizon is 48 hours. Hyperparameters were selected once on the 2023 validation block and held fixed. Three random seeds are used; the paired bootstrap resamples test weeks and uses the seed-average per week, so seed variance is not double-counted. About 24 paired comparisons are reported; we do not apply a multiple-comparison correction. The column labelled $P(A<B)$ in the bootstrap tables is the fraction of bootstrap replicates where A beats B, not a classical p-value. A Bonferroni threshold at $\alpha = 0.05$ would require $P<0.002$; the headline contrasts (wind-aware vs. distance for Graph WaveNet, and distance vs. no-graph for both architectures) satisfy that threshold except STGCN distance vs. no-graph ($P = 0.004$), which is treated as exploratory.

4.3 Meteorology

Weather is measured *at* each station rather than taken from a reanalysis, so the wind field carries genuine local structure: the mean cross-station correlation of the u-component is 0.11, and no two stations record the same wind in the same hour. This matters for the graph experiment—a wind-aware graph built on gridded reanalysis would have had almost nothing to add over a distance graph. It also introduces a caveat: a per-station *mean* wind direction estimated from weakly agreeing anemometers is itself a noisy quantity.

4.4 Graphs

Five structures are compared: **no graph** (identity, ablation baseline); **distance**, Gaussian kernel $\exp(-d^2/2\sigma^2)$ on great-circle distance, $\sigma = 5$ km; **correlation**, pairwise-complete Pearson correlation of PM2.5 on the training split, thresholded; **wind-aware**, a directed edge $i \to j$ only when the prevailing wind at i points towards j: $\exp(-d^2/2\sigma^2) \cdot \max(0, \cos(\theta_{ij} - \phi_i))$, a static graph oriented by the training-split period-mean direction; **learned adaptive**, $\mathrm{softmax}(\mathrm{relu}(E_1 E_2^\top))$, learned in-model.

The five structures are compared across all configurations in Table 4.

5 Results

5.1 Main comparison

Table 3: Test set (2024–2025), masked metrics in μg/m^3. Neural models averaged over 3 seeds.

Model	MAE	RMSE	R^2	1h	3h	6h	12h	24h	48h
stgcn_identity_netmean	**32.60** ± 0.08	57.27	0.626	15.06	23.82	28.52	31.03	32.91	36.39
stgcn_identity	**32.61** ± 0.08	57.28	0.626	15.05	23.82	28.52	31.04	32.93	36.38
lstm_univariate_distance	**32.80** ± 0.05	57.59	0.622	15.45	23.97	28.63	31.11	33.15	36.71
stgcn_wind_aware_era5	**32.92** ± 0.07	57.54	0.622	15.36	23.80	28.54	31.19	33.52	36.91
stgcn_wind_aware	**33.06** ± 0.10	57.72	0.620	15.59	23.96	28.80	31.30	33.68	36.96
stgcn_distance	**33.18** ± 0.08	58.18	0.614	15.40	24.15	28.78	31.54	33.71	36.99
stgcn_correlation	**33.47** ± 0.35	58.17	0.614	15.57	24.26	29.25	32.06	33.82	36.86
graph_wavenet_identity	**33.57** ± 0.15	58.50	0.610	15.36	24.37	29.22	31.91	33.63	37.55
graph_wavenet_wind_aware	**33.76** ± 0.25	58.92	0.604	15.53	24.18	29.13	31.90	33.80	37.67
graph_wavenet_wind_aware_era5	**33.95** ± 0.45	58.40	0.611	15.52	24.34	29.34	31.97	33.89	37.83
graph_wavenet_distance	**34.39** ± 0.03	59.45	0.597	15.66	24.36	29.42	32.44	34.49	38.48
graph_wavenet_learned_adaptive	**34.66** ± 0.19	59.86	0.591	16.02	25.18	29.92	32.71	34.70	38.27
graph_wavenet_correlation	**35.03** ± 0.48	60.10	0.588	15.96	25.14	30.64	32.58	35.00	38.61
lstm_multivariate_distance	**36.72** ± 0.74	65.63	0.508	16.13	25.46	30.94	34.17	37.19	41.36
Seasonal naive (24h)	38.63	68.16	0.470	38.86	38.56	38.39	38.58	38.41	38.60
transformer_distance	**38.81** ± 1.10	68.15	0.469	16.37	26.05	32.03	35.43	39.05	44.08
Ridge(24 lags)	44.03	73.69	0.380	15.83	28.09	39.46	47.14	36.45	42.34
Persistence	46.79	79.01	0.288	16.55	31.15	42.78	50.09	38.41	44.74
Hour-of-week average	59.86	95.39	−0.038	60.15	59.68	59.80	59.70	59.71	59.86

5.2 Graph ablation

Table 4: Graph structures compared under identical training and evaluation (3 seeds).

Graph	Model	MAE	RMSE	R^2
No graph	stgcn	32.61 ± 0.08	57.28	0.626
No graph	graph_wavenet	33.57 ± 0.15	58.50	0.610
Wind-aware	stgcn	33.06 ± 0.10	57.72	0.620
Wind-aware	graph_wavenet	33.76 ± 0.25	58.92	0.604
Wind-aware (ERA5)	stgcn	32.92 ± 0.07	57.54	0.622
Wind-aware (ERA5)	graph_wavenet	33.95 ± 0.45	58.40	0.611
Distance	stgcn	33.18 ± 0.08	58.18	0.614
Distance	graph_wavenet	34.39 ± 0.03	59.45	0.597
Correlation	stgcn	33.47 ± 0.35	58.17	0.614
Correlation	graph_wavenet	35.03 ± 0.48	60.10	0.588
Learned adaptive	graph_wavenet	34.66 ± 0.19	59.86	0.591

The correlation graph deserves a brief note. It is the second-worst graph for Graph WaveNet (35.03 ± 0.48, worse than the distance graph by 0.64) and also the highest-variance one. No bootstrap row is reported for it because the study's paired comparisons were specified in advance around the wind-aware hypothesis; the correlation graph is included as a reference point. Its high variance (± 0.48 over 3 seeds) suggests it is sensitive to the training draw, and the fact that it is beaten by every other graph type on both architectures is sufficient to set it aside.

5.3 Paired bootstrap

Table 5: Paired bootstrap over test weeks for the MAE difference (A minus B). Negative means A is better. The interval excludes zero when the difference is reliable.

comparison	ΔMAE	95% CI	$P(A < B)$	weeks
stgcn distance minus stgcn wind aware	+0.12	[−0.15, +0.41]	0.214	105
stgcn distance minus stgcn identity	+0.57	[+0.14, +1.01]	0.004	105
stgcn wind aware minus stgcn identity	+0.45	[+0.15, +0.79]	0.002	105
gwn distance minus gwn wind aware	+0.63	[+0.36, +0.91]	0.000	105
gwn distance minus gwn identity	+0.82	[+0.36, +1.30]	0.000	105
gwn wind aware minus gwn identity	+0.19	[−0.15, +0.55]	0.149	105
gwn learned adaptive minus gwn identity	+1.09	[+0.37, +1.87]	0.001	105
gwn learned adaptive minus gwn wind aware	+0.90	[+0.43, +1.39]	0.000	105
stgcn wind aware era5 minus stgcn wind aware	−0.14	[−0.35, +0.07]	0.904	105
stgcn wind aware era5 minus stgcn identity	+0.31	[−0.01, +0.64]	0.028	105
gwn wind aware era5 minus gwn wind aware	+0.19	[−0.07, +0.45]	0.080	105
gwn wind aware era5 minus gwn identity	+0.38	[−0.00, +0.78]	0.027	105
gwn correlation vs gwn identity	+1.46	[+0.70, +2.29]	1.000	105
stgcn correlation vs stgcn identity	+0.86	[+0.27, +1.49]	0.998	105
gwn correlation vs gwn distance	+0.62	[+0.15, +1.14]	0.997	105
stgcn identity minus lstm univariate	−0.19	[−0.33, −0.06]	0.999	105
stgcn identity minus Ridge(24 lags)	−11.42	[−13.18, −9.75]	1.000	105
stgcn identity minus Persistence	−14.18	[−16.24, −12.20]	1.000	105
stgcn identity minus Seasonal naive (24h)	−6.02	[−7.37, −4.82]	1.000	105
stgcn identity minus Hour-of-week average	−27.25	[−33.31, −21.57]	1.000	105
stgcn wind aware minus Seasonal naive (24h)	−5.57	[−6.87, −4.39]	1.000	105
gwn wind aware minus Seasonal naive (24h)	−4.87	[−6.13, −3.73]	1.000	105
lstm multivariate minus Seasonal naive (24h)	−1.91	[−3.60, −0.05]	0.977	105
stgcn identity netmean minus stgcn identity	−0.01	[−0.02, +0.00]	0.959	105
stgcn identity netmean minus stgcn wind aware	−0.46	[−0.79, −0.16]	0.999	105
stgcn identity netmean minus Seasonal naive	−6.03	[−7.41, −4.82]	1.000	105
stgcn identity netmean minus gwn identity	−0.97	[−1.20, −0.74]	1.000	105

5.4 The headline margin at each horizon

The best neural model beats the best simple baseline at every horizon. The per-horizon curve of the *baselines* is not monotone: the 12-hour column is worse than the 24-hour one for persistence and ridge. This is a property of Delhi's PM2.5 data, which has a diurnal cycle with a peak-to-trough amplitude of approximately 51 μg/m^3: a 12-hour horizon lands on the opposite phase while a 24-hour horizon returns to the same phase. Neural models, which learn the diurnal structure, do not show this non-monotonicity.

Table 6: Best neural model vs. strongest simple baseline at each horizon. Ridge is strongest at 1h, 3h, 24h; seasonal naive at 6h, 12h, 48h.

horizon	best neural	strongest simple	its MAE	margin
1h	15.05	Ridge	15.83	4.9%
3h	23.80	Ridge	28.09	15.3%
6h	28.52	Seasonal	38.39	25.7%
12h	31.03	Seasonal	38.58	19.6%
24h	32.91	Ridge	36.45	9.7%
48h	36.38	Seasonal	38.60	5.8%

5.5 Is the null result caused by the wind speed?

Windows binned by mean wind speed over the input period show that the no-graph variant is ahead in the two lower-wind thirds and essentially tied in the windiest third. The severe-episode analysis below explains why: severe episodes are disproportionately concentrated under calm conditions (78.2% of the lowest-wind third contains a severe-episode hour, against 52.1% and 49.0% in the windier thirds), and it is there that all graphs pay their largest penalty.

5.6 Is the result an artefact of the train/test boundary?

Because several real years are available, we also ran a rolling-origin evaluation: fold k trains on 2019 through year Y, validates on Q4 of Y for early stopping, and tests on year $Y+1$. The evaluation was run for STGCN (all five folds, test years 2021–2025) and Graph WaveNet (folds testing 2021, 2023, 2025). Note that two folds (testing 2024 and 2025) overlap with the main test block and are therefore not fully independent. The no-graph variant was the outright winner in three of five STGCN folds and within a few tenths in the rest. For Graph WaveNet the wind-aware graph won two of three folds, which is the opposite of the no-graph ordering on that architecture. The rolling-origin result is therefore best read as showing that the STGCN ranking is stable as the boundary moves, while the Graph WaveNet ranking is less so; the main test-block result is the more reliable comparison because it uses all three seeds and 105 weeks of data.

5.7 Case studies

Three real events from the test period are chosen by construction rather than by inspecting model wins.

Table 7: MAE during three real events (μg/m^3, network mean, 48-hour forecasts).

event	start	peak	persistence	ridge	no graph	wind-aware	GWN wind
Diwali 2024	2024-10-30	221	53.84	59.49	66.04	80.63	91.49
Winter inversion	2024-01-13	575	108.57	86.26	19.98	20.21	25.91
Dust storm	2024-05-10	162	15.48	15.62	17.93	15.60	14.10

The three events are informative in different ways. In the January inversion, a slow build-up the 72-hour window can partly see, the neural models are far ahead of both baselines. Here the STGCN no-graph (19.98) and STGCN wind-aware (20.21) are essentially tied, while GWN wind-aware (25.91) is notably worse. In the May dust episode, the wind-aware graph variants do better than no-graph, and GWN wind-aware (14.10) beats ridge (15.62) — the one case in the study where the wind-aware graph pays off against a linear baseline. Diwali is the hard case:

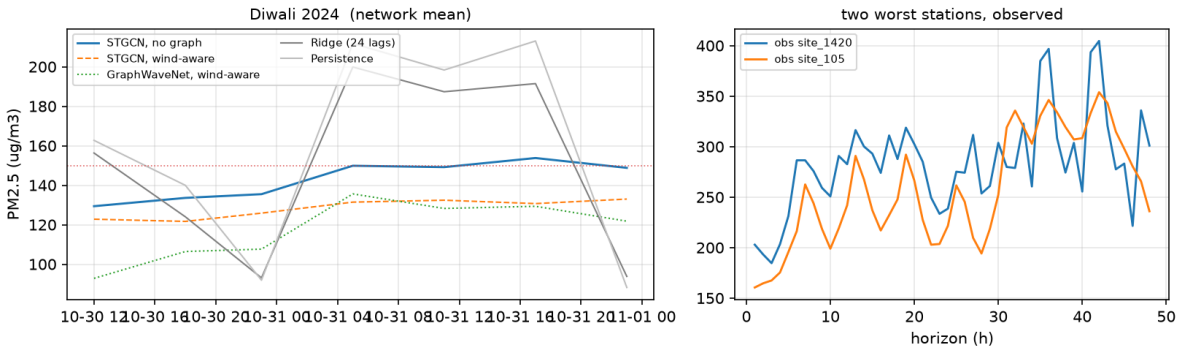


Figure 1: Diwali, 30–31 October 2024 (network mean + two worst stations). The fireworks spike is a human-caused event with no precursor in the meteorological inputs; neural models smooth over it and all lose to persistence and ridge.

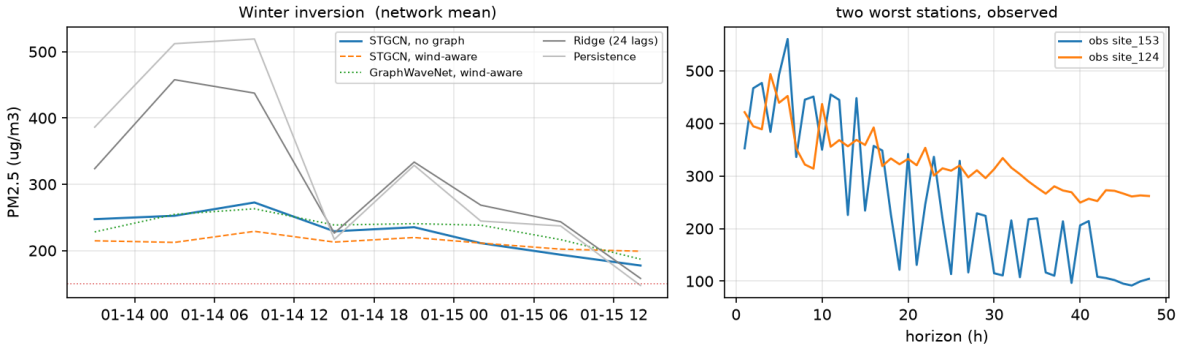


Figure 2: Winter inversion, 13–15 January 2024. A slow regional build-up that the 72-hour input window can partly anticipate; neural models are far ahead of both simple baselines.

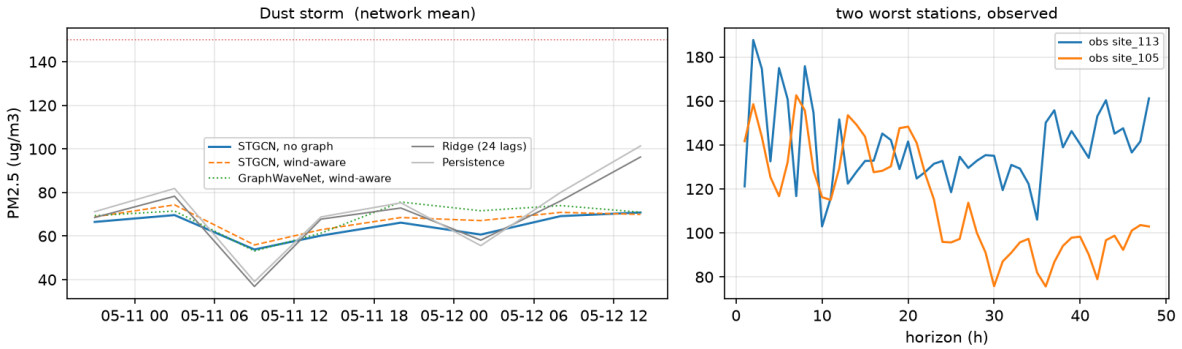


Figure 3: Dust episode, 10–12 May 2024. Well-mixed conditions; the wind-aware graph variants do better than the no-graph variant here.

a sharp anthropogenic spike with no precursor in the inputs; here neural models smooth over the spike and every neural variant loses to persistence and to ridge. Note that in the dust case the no-graph model (17.93) also loses to ridge (15.62), so the neural advantage is not universal.

5.8 Spatial error: who pays for the mixing?

If graph structure carried information a station cannot see, the benefit should be largest at the edge of the network. The evidence goes the other way. For STGCN, both graph types impose a larger penalty at peripheral stations (beyond the median centroid distance of 10.9 km) than at central ones: wind-aware +0.21 near vs. +0.72 far; distance +0.38 near vs. +0.78 far. The Spearman correlation between centroid radius and per-station penalty is +0.41 (wind) and +0.33 (distance). For Graph WaveNet the wind-aware pattern is similar (−0.10 near vs. +0.32 far, $r = +0.41$) but the distance graph has a slightly larger penalty near the centre (+0.89 near vs. +0.85 far, $r = +0.15$, indistinguishable from zero at 42 stations). The STGCN pattern is consistent with peripheral stations losing most from mixing; the Graph WaveNet distance result does not confirm this.

5.9 Adjacency introspection

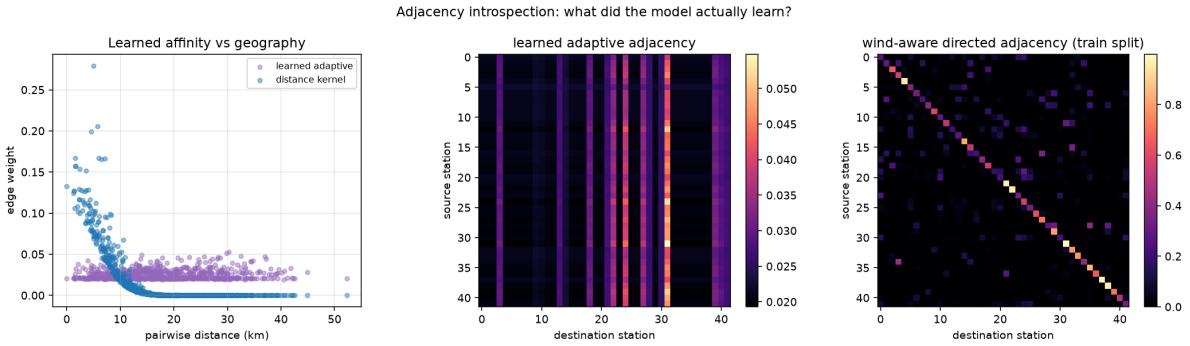


Figure 4: What the network learned: self-adaptive affinity against pairwise distance (Spearman $r = +0.07$) and against the wind alignment it was never shown ($r = -0.06$). The network neither rediscovers geography nor transport.

The self-adaptive adjacency that Graph WaveNet learns is essentially uncorrelated with pairwise distance (Spearman $r = +0.07$, against -1.00 for the distance kernel by construction) and with the prevailing-wind alignment ($r = -0.06$). The network learns a structure that is neither the distance graph nor the wind-aware graph, and that structure does not pay. The correlation graph and the learned adaptive graph are the two worst configurations. The correlation graph (35.03 μg/m^3) is the worst neural model overall, significantly worse than no graph for both architectures (STGCN: +0.86, CI [0.27, 1.49]; GWN: +1.46, CI [0.70, 2.29]) and worse than the distance graph for Graph WaveNet (+0.62, CI [0.15, 1.14]). The learned adaptive graph (+1.09 vs. no graph, CI [0.37, 1.87]) is second worst among GWN configurations.

5.10 Is the null result caused by the noisy local wind?

Nine of the 42 stations report no wind at all in the CPCB files. For those stations the wind-aware graph is built from the network-mean direction. A natural objection is that the edge directions are therefore noisy. ERA5 reanalysis is the opposite treatment: complete for every station, spatially smooth (cross-station correlation 0.98), but on a coarse grid. Replacing station wind with ERA5 wind, changing nothing else, isolates the wind source as the only variable.

The wind source does not matter. For STGCN, ERA5 wind and station wind are indistinguishable ($\Delta = -0.14$, CI $[-0.35, +0.07]$). For Graph WaveNet, the difference is $+0.19$ (CI $[-0.07, +0.45]$). Crucially, the ERA5 variants do not beat no graph either: STGCN ERA5 vs. no graph is $+0.31$ (CI $[-0.01, +0.64]$, $p = 0.028$, marginal) and GWN ERA5 vs. no graph is $+0.38$ (CI $[-0.00, +0.78]$, $p = 0.027$, also marginal). So neither wind source makes the graph significantly worse than no graph; but neither makes it significantly better.

The scope of this statement is narrow: it tests one edge construction against one alternative wind source. It does not show that wind cannot help. A time-lagged advection kernel, a wind field at higher spatial resolution, or gradient-driven edge weights are different constructions not tested here.

Table 8: ERA5-wind ablation: only the source of the wind changes.

model	edge direction	MAE	RMSE	R^2
STGCN	no graph	32.61 ± 0.08	57.28	0.626
STGCN	wind, station	33.06 ± 0.10	57.72	0.620
STGCN	wind, ERA5	32.92 ± 0.07	57.54	0.622
STGCN	distance	33.18 ± 0.08	58.18	0.614
GraphWaveNet	no graph	33.57 ± 0.15	58.50	0.610
GraphWaveNet	wind, station	33.76 ± 0.25	58.92	0.604
GraphWaveNet	wind, ERA5	33.95 ± 0.45	58.40	0.611
GraphWaveNet	distance	34.39 ± 0.03	59.45	0.597

5.11 Is the graph just delivering the regional signal?

We train a no-graph model with the network-wide mean PM2.5 as an extra input channel and compare. It buys essentially nothing: 32.60 against 32.61 without it, a paired difference of 0.01 μg/m^3 (CI $[-0.02, +0.00]$), 0.03% of the score. And the same model is significantly better than every graph model (0.46 better than STGCN wind-aware, CI [0.16, 0.79]; 0.97 better than GWN without a graph, CI [0.74, 1.20]). The graph is not delivering the regional signal and being ignored for it; delivering the regional signal explicitly is worth essentially nothing.

Table 9: Network-mean diagnostic.

configuration	MAE	RMSE	R^2
stgcn, no graph	32.61 ± 0.08	57.28	0.626
stgcn, no graph + network mean	32.60 ± 0.08	57.27	0.626
stgcn, wind-aware	33.06 ± 0.10	57.72	0.620
gwn, no graph	33.57 ± 0.15	58.50	0.610
gwn, wind-aware	33.76 ± 0.25	58.92	0.604

5.12 Can a graph help a station the model has never seen?

Ten stations are held out entirely from training. Their targets never enter the training loss. Two evaluation variants are compared: *mode target*: the station's own 72-hour history is still visible but its targets are hidden; *mode both*: both the targets and the station's own history are hidden, so the only information available comes from neighbours.

In mode target the picture is the same as the main experiment: the no-graph variant is best (STGCN identity 35.35 μg/m^3) and both graph variants are worse, though with wide confidence

intervals (wind vs. no-graph: +0.39, CI [−0.04, +0.85]; distance vs. no-graph: +0.59, CI [−0.01, +1.22]). These models do generalise to new monitors: STGCN identity beats a ridge fitted on other stations by 12.1 μg/m^3 and seasonal naive by 6.1.

In mode both the honest comparator is not the no-graph model (which has nothing to work with and collapses, $R^2 = -0.32$) but the simplest spatial baselines that use the same information the graph has. An inverse-distance-weighted average of the neighbours' values from 24 hours before the target (IDW-seasonal, 41.24 μg/m^3, $R^2 = 0.51$) beats the best graph model (STGCN distance, 54.32 μg/m^3) by 13.1 μg/m^3 (CI [8.69, 18.04]). This comparison has a known handicap: these models were trained to forecast a station *from its own history* and are now asked to forecast one whose history has been removed—a condition they never practised, while the IDW baseline has no such dependence. The honest claim is therefore the narrow one: graph models trained for standard forecasting do not transfer to missing-history stations as well as a two-line distance weighting.

Table 10: Leave-station-out: MAE on held-out stations only (μg/m^3). Neural models never saw those stations' targets; ridge fitted on training stations only. Two seeds per configuration.

configuration	MAE	R^2	1h	24h	48h
Mode both — held out: site_106, site_107, site_108, site_1422, site_1427, site_1432, site_1434, site_1435, site_1561, site_5024					
Persistence	50.73	0.269	17.35	42.04	47.79
Seasonal naive (24h)	41.40	0.479	41.47	42.04	41.52
Hour-of-week average	63.48	−0.052	63.97	64.11	64.10
Ridge (fit on training stations only)	47.48	0.378	16.66	39.52	44.79
Network mean (persistence)	47.01	0.402	28.45	40.83	45.11
IDW neighbour (persistence)	47.65	0.376	28.77	41.86	46.28
IDW neighbour (seasonal)	41.24	0.506	41.39	41.77	41.31
stgcn_distance_s123_lso10_both	53.42	0.147	62.06	54.31	52.56
stgcn_distance_s42_lso10_both	55.27	0.111	62.42	56.73	55.28
stgcn_wind_aware_s123_lso10_both	55.22	0.118	63.49	56.06	53.46
stgcn_wind_aware_s42_lso10_both	55.99	0.103	63.44	57.31	55.66
gwn_distance_s123_lso10_both	54.10	0.124	60.82	55.29	55.18
gwn_distance_s42_lso10_both	57.16	0.044	61.93	58.75	58.62
gwn_wind_aware_s123_lso10_both	56.81	0.048	61.35	57.51	55.97
gwn_wind_aware_s42_lso10_both	58.49	0.003	63.02	59.84	59.38
stgcn_identity_s123_lso10_both	69.94	−0.372	67.14	71.12	70.78
stgcn_identity_s42_lso10_both	68.59	−0.324	66.11	69.75	69.87
gwn_identity_s123_lso10_both	71.35	−0.415	77.60	72.99	70.47
gwn_identity_s42_lso10_both	69.40	−0.353	75.17	70.05	68.94
Mode target — held out: same 10 stations, own history visible					
Persistence	50.73	0.269	17.35	42.04	47.79
Seasonal naive (24h)	41.40	0.479	41.47	42.04	41.52
IDW neighbour (seasonal)	41.24	0.506	41.39	41.77	41.31
Ridge (fit on training stations only)	47.48	0.378	16.66	39.52	44.79
Network mean (persistence)	47.01	0.402	28.45	40.83	45.11
stgcn_identity_s123_lso10_target	35.33	0.614	15.75	35.76	39.35
stgcn_identity_s42_lso10_target	35.36	0.611	16.15	35.90	39.58
stgcn_wind_aware_s123_lso10_target	35.69	0.600	16.17	36.03	39.52
stgcn_wind_aware_s42_lso10_target	35.78	0.597	16.23	36.18	40.06
stgcn_distance_s123_lso10_target	35.79	0.589	16.19	36.43	39.63
stgcn_distance_s42_lso10_target	36.07	0.578	16.14	36.58	41.09
gwn_identity_s123_lso10_target	35.66	0.610	15.79	35.95	39.61
gwn_identity_s42_lso10_target	35.60	0.617	16.24	36.08	39.22
gwn_wind_aware_s123_lso10_target	36.36	0.602	16.06	36.72	40.48
gwn_wind_aware_s42_lso10_target	36.10	0.592	16.35	36.44	41.12
gwn_distance_s123_lso10_target	36.87	0.592	16.10	37.86	41.17
gwn_distance_s42_lso10_target	36.32	0.605	16.50	36.64	40.09

5.13 Severe episodes

Episodes are sustained periods with PM2.5 above 150 μg/m^3 for at least 6 hours. The network-wide episode-hour fraction is 15.6%.

Every architecture degrades sharply during episodes: episode error is roughly 3.3× the non-episode error (e.g. STGCN identity: 78.84 vs. 23.60 μg/m^3). Although episode hours are only 15.6% of test hours, they account for approximately 38% of the total absolute error, so the error budget is disproportionately dominated by episodes. Outside episodes the spread between

Table 11: Leave-station-out paired bootstrap (A minus B; negative = A better). Rows 1–9 are *mode target* (own history visible). Rows 10–18 are *mode both* (own history hidden; fair comparators: spatial baselines).

comparison	ΔMAE	95% CI	$P(A < B)$
Mode target (own history visible):			
stgcn wind aware vs stgcn identity	+0.39	[−0.04, +0.85]	0.038
stgcn distance vs stgcn identity	+0.59	[−0.01, +1.22]	0.026
gwn wind aware vs gwn identity	+0.60	[+0.09, +1.15]	0.010
gwn distance vs gwn identity	+0.97	[+0.41, +1.53]	0.000
stgcn identity vs ridge	−12.13	[−14.11, −10.27]	1.000
stgcn identity vs seasonal naive	−6.06	[−7.53, −4.76]	1.000
stgcn distance vs idw seasonal	−5.31	[−6.54, −4.14]	1.000
stgcn identity vs idw seasonal	−5.90	[−7.14, −4.77]	1.000
stgcn identity vs network mean	−11.67	[−13.61, −9.79]	1.000
Mode both (own history hidden; fair comparators: spatial baselines):			
stgcn distance vs stgcn identity	−14.92	[−18.78, −11.12]	1.000
stgcn wind aware vs stgcn identity	−13.66	[−17.57, −9.87]	1.000
gwn distance vs gwn identity	−14.75	[−18.51, −11.13]	1.000
gwn wind aware vs gwn identity	−12.72	[−16.05, −9.64]	1.000
stgcn distance vs idw seasonal	+13.10	[+8.69, +18.04]	0.000
stgcn distance vs idw persistence	+6.69	[+2.75, +11.15]	0.000
stgcn distance vs network mean	+7.33	[+3.33, +11.75]	0.000
gwn distance vs idw seasonal	+14.39	[+9.57, +19.77]	0.000
idw seasonal vs network mean	−5.77	[−7.39, −4.24]	1.000

Table 12: Test performance inside and outside severe episodes (μg/m^3). Episode error is roughly 2–3× non-episode error for every architecture.

Model	Overall MAE	Episode MAE	Non-episode MAE
stgcn_identity_netmean	32.65	78.78	23.60
stgcn_identity	32.65	78.84	23.60
stgcn_wind_aware	33.12	78.91	24.14
stgcn_distance	33.24	80.07	24.06
graph_wavenet_identity	33.69	80.50	24.52
graph_wavenet_wind_aware	33.81	81.62	24.44
graph_wavenet_distance	34.56	82.45	25.18
Ridge(24 lags)	43.97	98.10	33.36
Persistence	46.65	102.57	35.69

configurations is modest: the non-episode MAE ranges from 23.60 (STGCN identity) to 25.18 (GWN distance), a gap of 1.6 μg/m^3.

Note that the per-seed episode MAEs in this table are computed on 1,200 evenly-spaced windows and the seed-averaging is approximate, so the "overall MAE" column here (32.65) differs slightly from the 2,000-window figure in Table 3 (32.61); all comparisons should use Table 3.

6 Discussion

6.1 What the graph comparison shows

All five graph structures were trained and evaluated identically, with 3 seeds, and judged with a paired bootstrap over the 105 ISO weeks of the test period. The differences are statistically clear but small in absolute terms: between graph structures the gap reaches 1.1–1.5 μg/m^3 (3–4.5%), and the range from best (no graph, 32.61) to worst configuration (GWN correlation, 35.03) is 2.4 μg/m^3 (7%).

The picture is specific rather than a blanket negative. For Graph WaveNet the wind-aware graph is significantly better than the static distance graph: 33.76 against 34.39 μg/m^3, difference 0.63, CI [0.36, 0.91]. For STGCN the same comparison is inconclusive: 33.06 against 33.18, difference 0.12, CI [−0.15, 0.41].

What holds for both architectures is that every graph type is significantly worse than no graph, or at best indistinguishable from it. The distance graph is significantly worse (STGCN +0.57, CI [0.14, 1.01]; GWN +0.82, CI [0.36, 1.30]). The wind-aware graph is also significantly worse than no graph for STGCN (+0.45, CI [0.15, 0.79]); for Graph WaveNet the difference (+0.19, CI [−0.15, 0.55]) is not significant. The correlation graph is the worst configuration overall (GWN 35.03, +1.46 vs. no graph, CI [0.70, 2.29]; STGCN 33.47, +0.86, CI [0.27, 1.49]). The learned adaptive adjacency is the second worst GWN configuration (+1.09, CI [0.37, 1.87]).

The ERA5 ablation shows the wind source is not the issue. Neither wind source makes the graph significantly worse than no graph (both CIs include zero at the margins), and neither makes it significantly better. The null result is not an artefact of the noisy or missing local wind.

Two diagnostics sharpen the interpretation. The network-mean diagnostic shows the graph is not delivering the regional signal (0.01 μg/m^3 explicitly vs. the graph's mixing cost of 0.19–0.82). The leave-station-out result shows that in the regime designed to favour graph structure—a station with no history of its own—a graph neural network trained without node dropout does not match simple spatial interpolation; it is 13.1 μg/m^3 worse than IDW-seasonal.

Our answer to the research question is narrow but defensible. A wind-aware graph beats a static distance graph for Graph WaveNet and is indistinguishable from it for STGCN. In both architectures, every graph type is either significantly worse than no graph or indistinguishable from it; none is significantly better. The wind-aware graph is significantly worse than no graph for STGCN (0.45, CI [0.15, 0.79]) and not significantly different for Graph WaveNet (0.19, CI [−0.15, 0.55]). The correlation graph and the learned adaptive adjacency are the two worst configurations. Two diagnostics say why: the regional-signal diagnostic shows the graph's mixing cost (0.19–1.46 μg/m^3 across graph types and architectures) is not delivering a useful signal; and the leave-station-out result shows the graph trained without node dropout cannot match simple spatial interpolation in the missing-history regime.

6.2 Failure analysis

Three failure modes account for most of the residual error. First, abrupt anthropogenic spikes (Diwali, crop-residue burning) are not predictable from meteorology or neighbouring stations, and every model smooths them away. Second, sensor outages produce stale readings that the

mask channel at least flags. Third, severe episodes are under-predicted: the model's autoregressive signal is a poor guide when concentrations are building non-linearly, which is why episode error is roughly 3.3× the non-episode error for every architecture.

6.3 Limitations

- The test block covers two winters; generalisation across pollution regimes is only partly tested.
- The wind-aware graph is *static*: it uses a period-mean prevailing direction, not a time-varying one. A dynamic wind-aware graph is a different construction not tested here.
- The distance kernel uses $\sigma = 5$ km against a median station spacing of ∼16 km, which makes the kernel nearly diagonal. A larger-σ or k-nearest-neighbour distance graph is a different construction.
- Nine of the 42 stations report no wind at all. The ERA5 ablation shows this does not drive the result, but it limits the precision of the station-specific wind direction used to orient the graph.
- "Dilution of the autoregressive signal" is a hypothesis consistent with the ablation, the network-mean diagnostic, and the leave-station-out result, but is not directly demonstrated.
- Single region, single hazard; results may not transfer to other cities or pollutants.
- Graph structures are static except Graph WaveNet's learned variant.
- Leave-station-out models were trained without node dropout; the missing-history comparison therefore contains a known handicap.

7 Conclusion

Across seven years of real observations at 42 stations, every neural model we trained beats ridge regression and persistence. Most beat seasonal naive; the multivariate LSTM is modestly better (-1.91 μg/m^3, CI [-3.60, -0.05], about 5%), and the Transformer is worse ($+0.18$). The best model beats the best simple baseline by 15.6% in aggregate (32.61 vs. 38.63, CI [4.82, 7.37]) and by a positive margin at every horizon.

The graph question has a more specific answer than a simple negative. A wind-aware directed graph beats a static distance graph for Graph WaveNet (0.63 μg/m^3, CI [0.36, 0.91] by paired bootstrap) and is indistinguishable from it for STGCN (0.12, CI [-0.15, 0.41]). In both architectures, every graph type is either significantly worse than no graph or indistinguishable from it. The wind-aware graph is significantly worse than no graph for STGCN (0.45, CI [0.15, 0.79]) and not significantly different for Graph WaveNet (0.19, CI [-0.15, 0.55]). The correlation graph (35.03) and the learned adaptive adjacency (34.66) are the two worst GWN configurations; the correlation graph is the worst neural model overall.

Two diagnostics explain why. Handing a no-graph model the network-wide mean PM2.5 gains essentially nothing (0.01 μg/m^3), so the graph is not delivering a useful regional signal. And on a station the model has never seen and whose history is missing, a graph neural network trained without node dropout is 13.1 μg/m^3 worse than inverse-distance-weighted interpolation — not because the graph is useless, but because the models were trained for the regime in which a station's own history is available.

The consistent story is that the per-station autoregressive signal is what matters, and inter-station mixing at this station density provides at best a small benefit (wind-aware over distance

for Graph WaveNet) and at worst a net cost (distance graph in both architectures). The remaining error is concentrated almost entirely in severe pollution episodes. Whether a differently constructed graph pays off on a denser network, a finer temporal resolution, or a longer and more diverse dataset is left open.

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